

Meeting – June 11

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Start of the Week

- Last week, I stated the plan was to deploy a small mesh around campus and run tests on them, analyzing signal strength, quality and comparing efficiency between firmware.
- I was going to jump in and assume that under the the same settings, zero-hop communication with Meshtastic and MeshCore are essentially equal. However, I wanted some evidence to back this claim and decided run a simple test for my hypothesis.
- The test consisted of biking down University Ave (to Sexton and back), comparing range, RSSI and SNR between the two firmware.
- Unfortunately, this test is no longer relevant as I now have the resources to do the campus deployment solely with T-Beams.

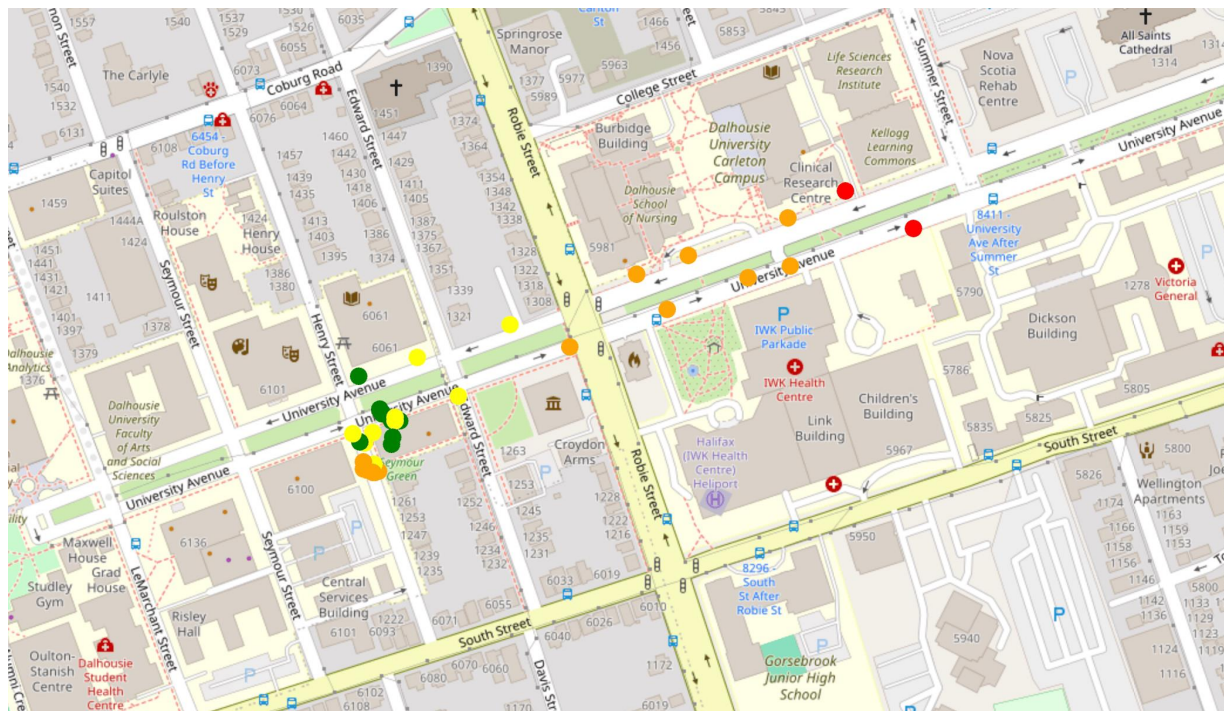
LoRa Settings

- *All the settings below were applied on both firmware.*
- Frequency: 906.875 MHz
- Bandwidth: 250 kHz
- Spreading Factor: 11
- Coding Rate: 5 (or 4/5)
- Transmit Power: 20 dBm

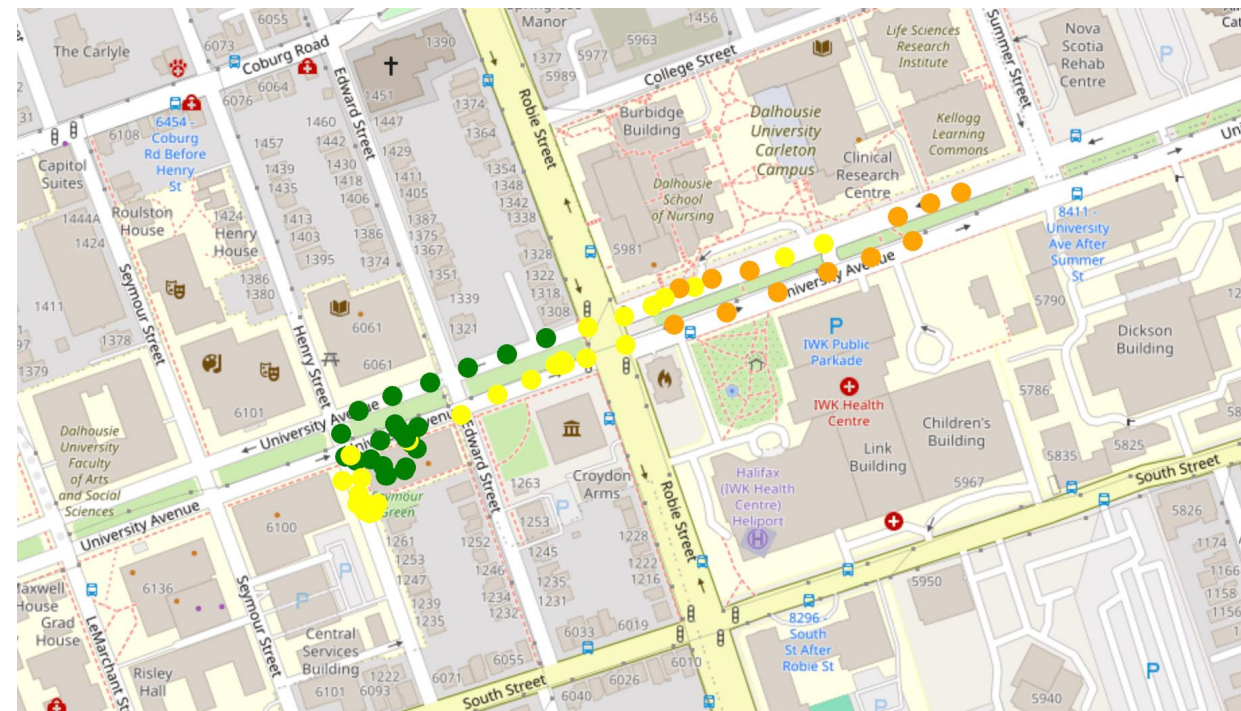
Basic Firmware Comparison Test Results

RSSI

Meshtastic



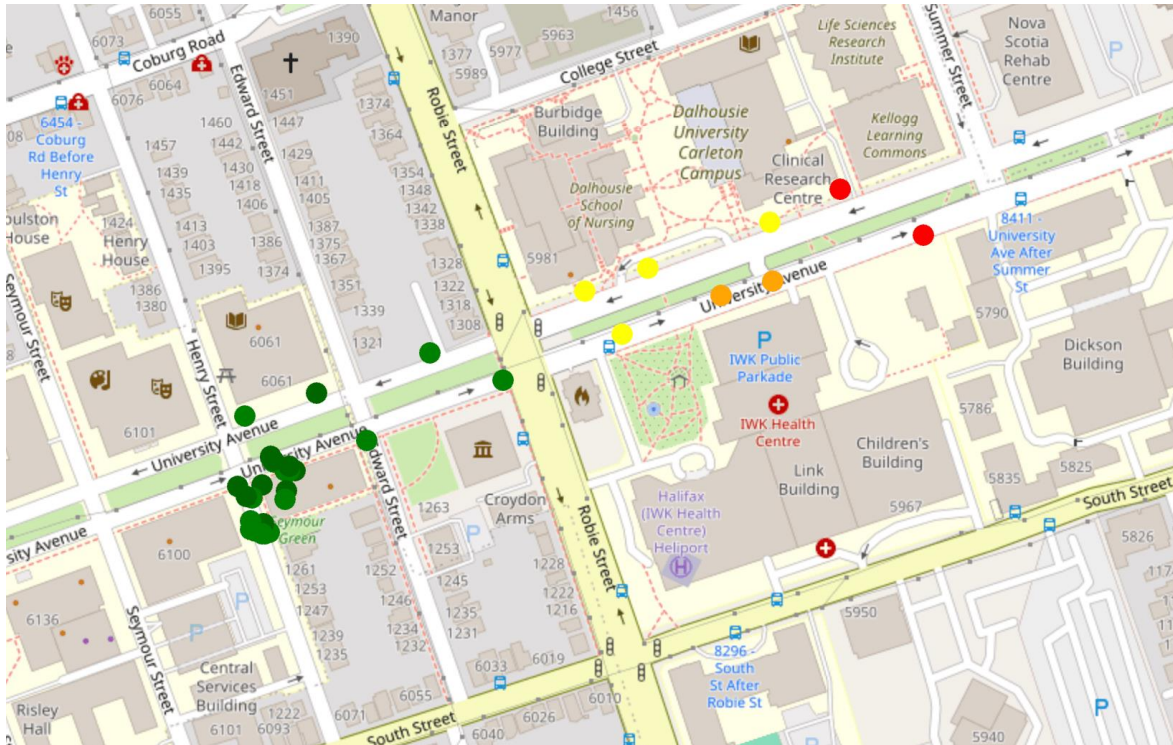
MeshCore



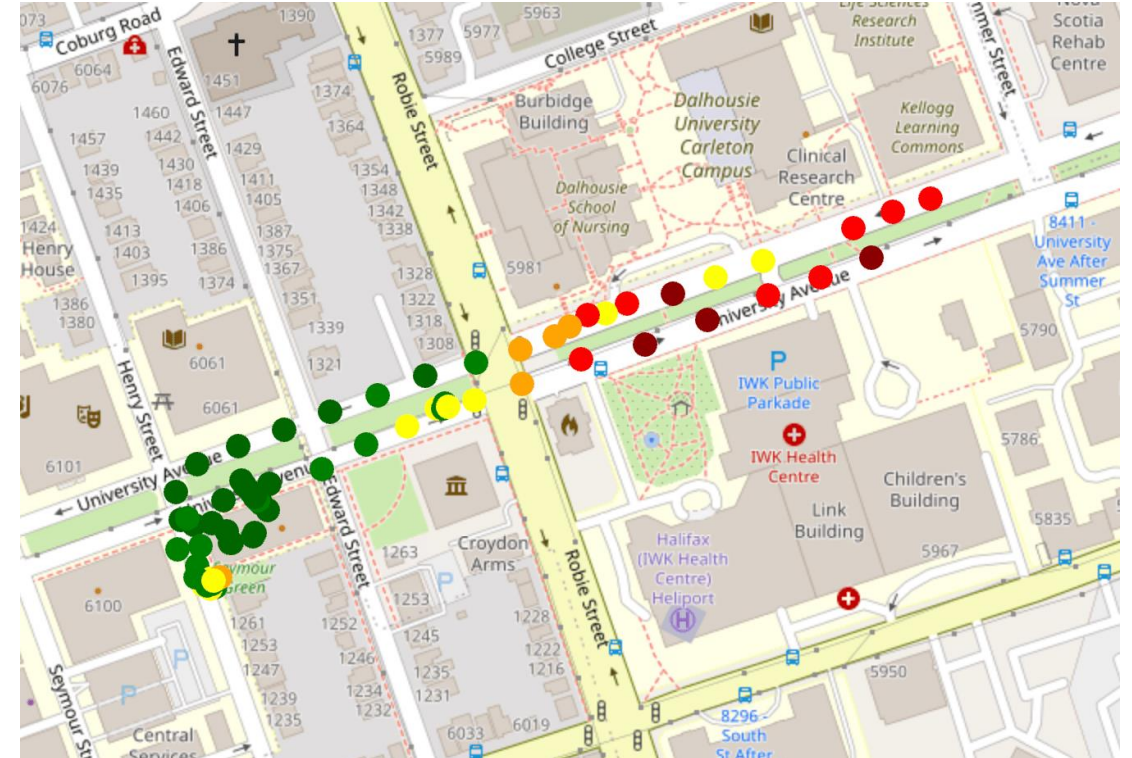
Basic Firmware Comparison Test Results

SNR

Meshtastic



MeshCore

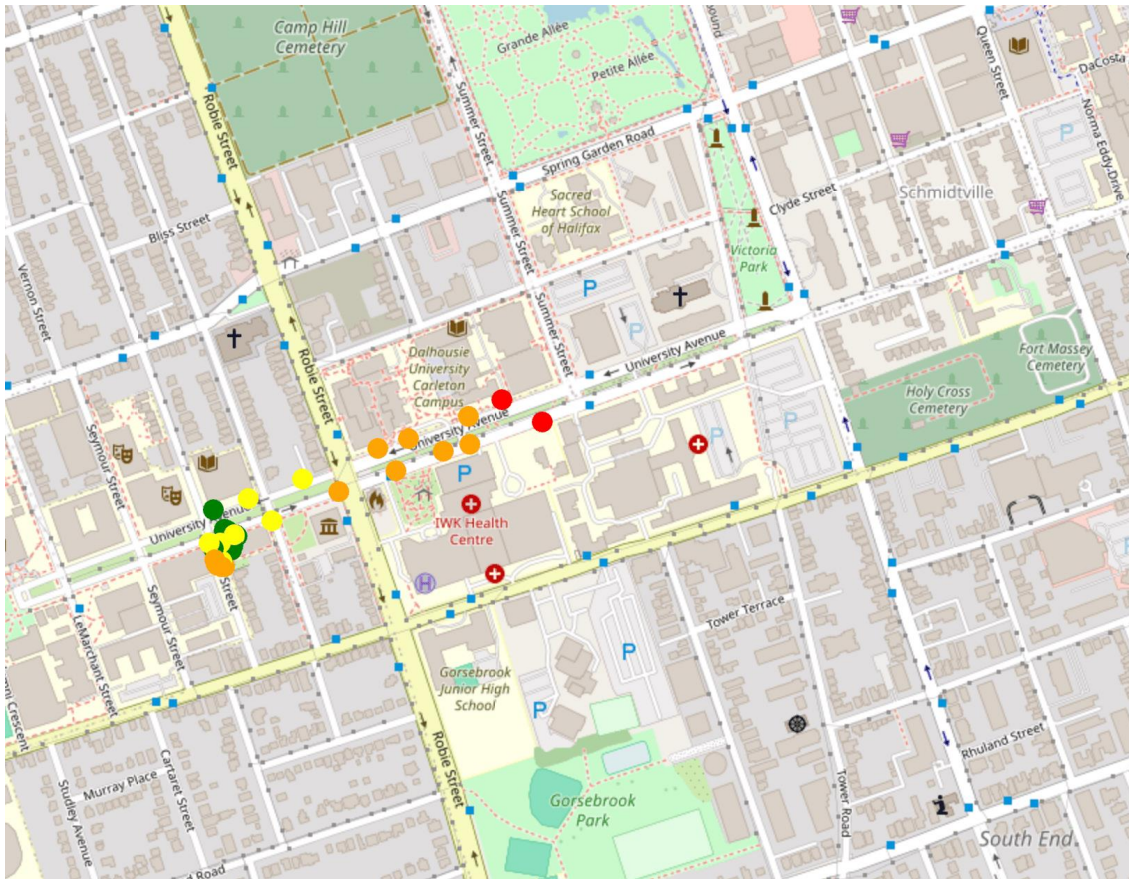


Results

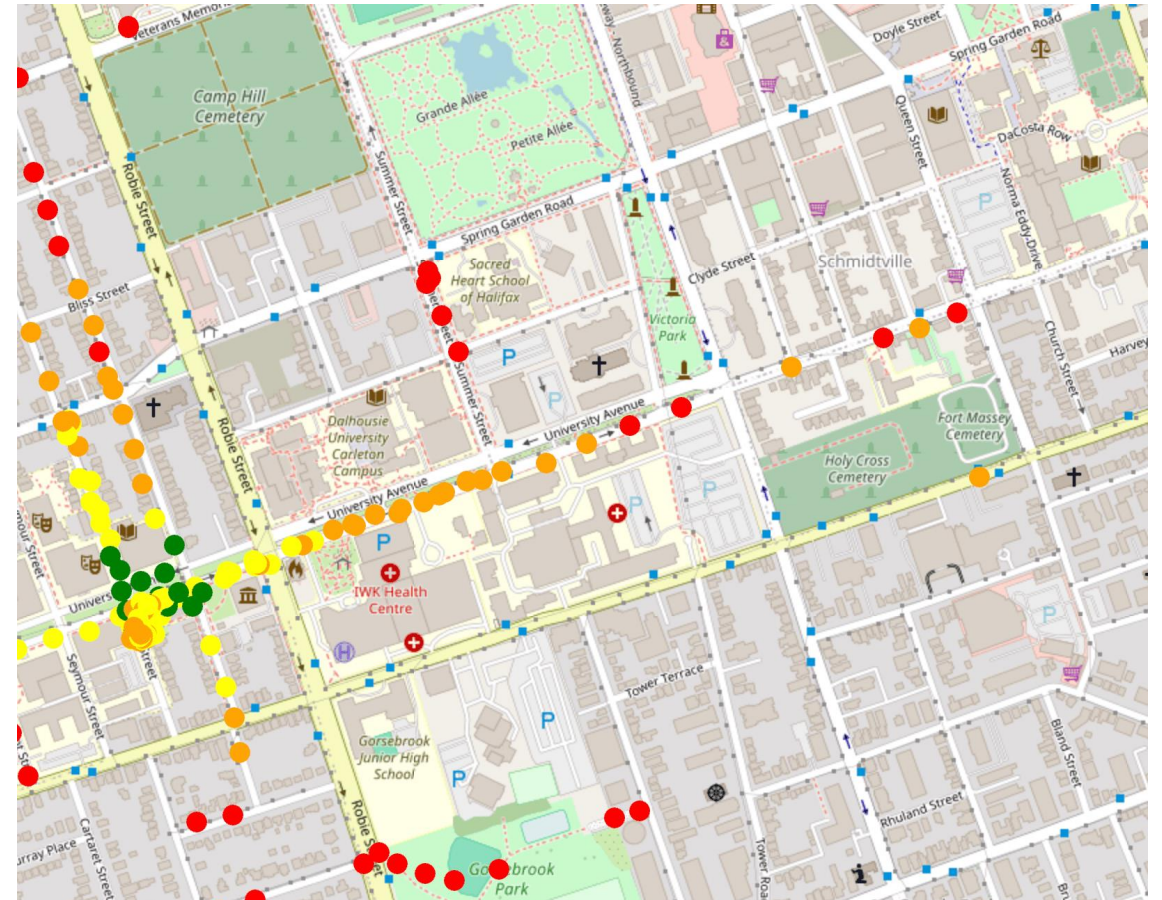
- Comparing the two firmware, there seems to be very little difference in all metrics, including range, RSSI and SNR.
- I believe the tests show prove enough to say that the two firmware perform similarly enough that further tests can be done under the assumption that they are equal under the same LoRa settings.
- The bigger takeaway, however, the range of this test compared to the previous bike tests. In these tests, I switched the sender node to the T3 LoRa32 V1.6.1, which has a much smaller default antenna.
 - I believe that this significant decrease in sensitivity can be attributed to the change in antenna.

Range Comparison

Small Antenna - T3 LoRa32 V1.6.1

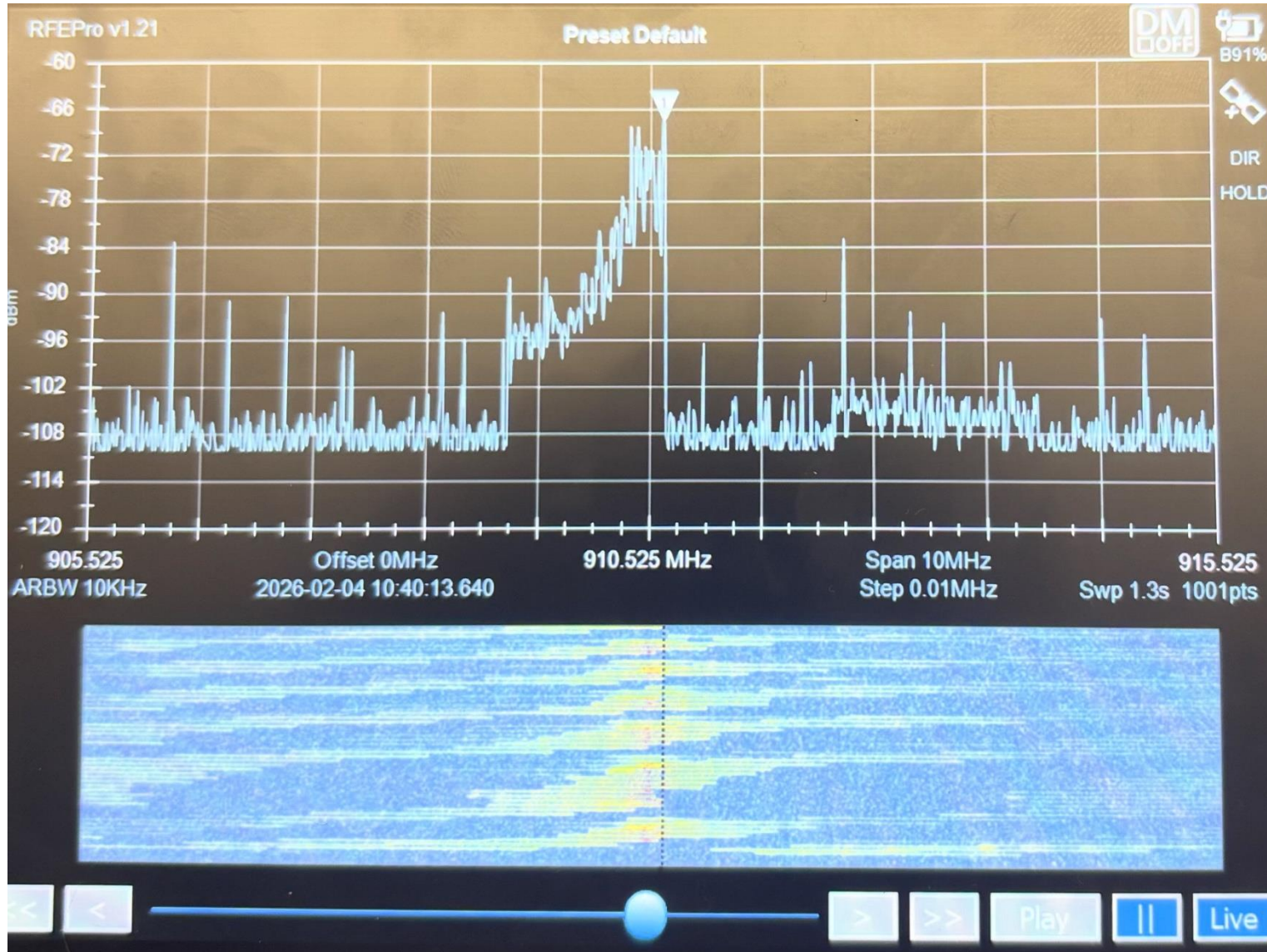


Large Antenna - T-Beam V1.1



RF Explorer Pro

- The RF Explorer Pro is an advanced instrument used to visualize and monitor a wide range of frequencies. The RF Explorer Pro has a variety of different tools but is often used for spectrum analysis.
- I tested the RF Explorer Pro by monitoring packets sent every two seconds between MeshCore nodes. I used the spectrum analysis tool, specifically focusing on the behavior of the waterfall graph.
 - Test was successful and as expected – see next slide
- In terms of antenna testing, the RF Explorer Pro can be used to measure the VSWR and Return Loss with additional hardware (signal generator, return loss bridge)
 - VSWR = How efficiently RF power is transmitted from source to antenna
 - Return Loss = essentially another way to measure VSWR



- Radio settings:
 - Freq = 901.525 MHz
 - BW = 62.5 kHz
 - SF = 8
 - CR = 5
- The LoRa chirps are easily observable, showing activity going from the bottom left to the top right.
- The most activity is concentrated around 910.515 MHz, as expected given the radio settings.

Campus Deployment

- The goal is to obtain concrete data comparing the routing efficiency and behavior between Meshtastic and MeshCore in an urban deployment.
 - Conclude on which firmware is better for an urban deployment and in what use cases is a mesh feasible or not feasible.
- The campus deployment should prove to be quite useful as there are several potential tests which could prove to be informative including:
 - Mesh signal strength heatmap
 - Routing comparison test (mobile and stationary)
 - Mesh stress test
 - Power consumption test

Potential Campus Deployment Locations



Not all these locations will be used, but as many as possible based on the amount of hardware and space to place nodes.

- Goldberg CS Building
- Student Union Building
- Mona Campbell Building
- Killam Library
- Marion McCain Building
- Studley Gym
- Sir James Dunn Building
- LSC (Psychology Tower)
- Dalplex
- *Maybe extend to Carlton Campus?*

Tests for Campus Deployment

- Initial mesh health test
 - Only on Meshtastic – using the node-neighbor module to graph out the mesh
- Routing tests
 - Stationary test: Sending packets through the mesh at different point, comparing hops, routes taken and the approximate latency
 - Mobile test: Moving node will send packets to gateway at center of mesh. Compare how each firmware's routing system adapts to changes in node position
- Battery/power consumption tests
 - Periodically sending messages through the mesh and requesting telemetry to gather mesh health over time. Would be good to do this over an extended period (overnight)

Currently

- Writing all the code required for both Meshtastic and MeshCore to do these tests.
- Everything is ready to go except for the battery/power consumption tests.
- Gathering enough wall to USB-A plugs such that campus deployment can begin tomorrow.
- Obtaining 18650 batteries required for the T-Beams
 - Canadian Tire sells these batteries. Still need to find out where to get an 18650 battery charger.

Next Week

- Conduct all the above tests and analyze the results!
- Aside: I will likely do all the tests on one firmware first then switch as going around campus and re-flashing every node will be extremely time consuming.